

The AUD/JPY triangular basis

Estimation and reconciliation at one-second resolution

AUD/JPY against $\text{AUD/USD} \times \text{USD/JPY}$

1 July to 30 August 2024

Scope. The triangular basis is the residual of the parity relation between AUD/JPY, AUD/USD and USD/JPY. This report documents the estimators used to measure that residual at one-second resolution over a window containing the August 2024 yen carry unwind, and reconciles the estimates that disagree. The residual remains inside the transaction-cost band throughout the sample; no arbitrage is identified and none is claimed.

Principal result. Basis dispersion over the stress window is 3.09 times its level in the calm segments when every second is pooled, and 1.22 times when measured within the hour. Section 7 shows that the difference is a variance decomposition rather than a modelling disagreement. Day-to-day variation in the basis level accounts for 80.6% of the stress window's level variance, against 11.6% in the calm segments, and that split holds across every candidate regime boundary. Two days in that window fail the daily-bias check of Section 2 by a factor of fifteen and account for the entire mean level shift attributed to the regime.

Results independent of that decomposition. Hourly dispersion is best described by a bounded excursion rather than a permanent shift, with posterior mass 1.000 on the two-changepoint models and an observed log Bayes factor of +97.1 against a null 95th percentile of -2.8 (Section 3). The transaction-cost band implied by quoted spreads exceeds the median basis by a factor of 12.7 at its tightest (Section 6). The two dollar legs open the observed dislocations and the cross leg closes them, established both by arithmetic decomposition and under a sign restriction fixed before estimation (Sections 4 and 5).

The emphasis throughout is on how each quantity is defined and why it is identified. Every number cited is reproduced from a table written by the pipeline. The appendix maps each claim to its file.

1 The estimand

1.1 The parity relation

Let S_t^{AU} , S_t^{UJ} and S_t^{AJ} denote mid quotes for AUD/USD, USD/JPY and AUD/JPY. Absent frictions the cross rate is redundant, $S_t^{\text{AJ}} = S_t^{\text{AU}} S_t^{\text{UJ}}$. The object of study is the residual of that relation, expressed in AUD/JPY pips:

$$z_t = \frac{S_t^{\text{AJ}} - S_t^{\text{AU}} S_t^{\text{UJ}}}{10^{-2}}. \quad (1)$$

For the dynamic work the same quantity is written in logs. With $x_t = (x_t^{\text{AU}}, x_t^{\text{UJ}}, x_t^{\text{AJ}})^\top$ the vector of log prices scaled by 10^4 and

$$w = (-1, -1, +1)^\top, \quad \tilde{z}_t = w^\top x_t, \quad (2)$$

$\tilde{z}_t$ agrees with (1) to first order. The ratio between the two scales is measured at load rather than assumed, and holds to about two percent at the sample average of the cross.

1.2 The cointegrating vector is known

Triangular parity is a cointegrating restriction on x_t with a known vector. The rank is one and the vector is w , both by arithmetic. Nothing about the relation is recovered from the data, which removes the Johansen procedure and the question of how many cointegrating relations the sample supports. Every dynamic statement below is therefore conditional on a restriction that cannot be misspecified, and any failure to find mean reversion is a statement about the market rather than about the specification.

1.3 Three distinct estimands

The phrase “the basis widened” names at least three quantities. They are defined here and returned to in Section 7.

1. **Within-hour dispersion.** $\text{MAD}(z_t : t \in h)$ for an hour h . This is the feature the change-point model of Section 3 fits.
2. **Within-day dispersion.** $\mathbb{E}_d[\text{Var}(z \mid d)]^{1/2}$, which pools hours within a day but not days.
3. **Pooled dispersion.** $\text{Var}(z)^{1/2}$ over every second in a regime, which additionally absorbs the movement of the daily mean.

The three coincide only if the daily mean basis is constant. It is not, and under stress it is far from constant. That fact accounts for most of the disagreement between sections of this project.

1.4 Episodes

Dislocations in this sample are plateaus rather than isolated spikes, so several statements are made at the level of an episode: a maximal run of consecutive seconds with $|z_t| > u$, where $u = 2.42$ pips is the 99.95% quantile of $|z|$ over July. Runs are split wherever consecutive observations are more than one grid step apart, which prevents an episode from spanning a market closure. Without that guard, exceedances either side of a weekend merge into a single 48-hour episode. Fifty-eight episodes are detected.

2 Constructing the observation grid

2.1 The failure a tick audit cannot detect

The ingest stage audits every tick that exists: nulls, zeros, negatives, crossed quotes. That audit is silent on ticks that are absent, because a missing tick leaves no row to inspect. Forward-filling then substitutes a stale price, which is correct when a quote is late and wrong when a feed is dead. Nothing in the filled series distinguishes the two cases.

Two design decisions follow. Every row carries an explicit freshness flag per leg, taken from whether that second contained a genuine tick rather than inferred from whether the price changed, since a repeated identical quote is fresh and a difference cannot detect that. And a separate gate runs between ingest and analysis.

2.2 Two statistics from the over-identification

Each leg is implied by the other two, for instance $\widehat{S^{UJ}} = S^{AJ} / S^{AU}$, and the implied series must agree with the quoted one. Two statistics are taken from that gap. They detect different failures and neither detects the other's.

- The 99.9th percentile of the *absolute* gap, a tail statistic sensitive to a frozen or wildly wrong leg. Observed: 3.21, 7.29 and 4.83 pips against a threshold of 25.
- The *median signed* gap within each day. A leg biased by a fraction of a pip moves this immediately and does not move the tail statistic at all, because the tail is set by ordinary noise an order of magnitude larger.

The second failure is the more consequential, because everything downstream continues to look plausible and any analysis that differences, measures dispersion, or centres per day removes the bias before it becomes visible. Section 8 reports what this check returned.

Closures are excluded from both statistics. Across a weekend the grid is forward-filled padding, so roughly 172,000 identical values carry an identical frozen gap. Computed on the full grid, the 99th and 99.9th percentiles were bit-identical for all three legs and both equal to a single stale value.

2.3 Closures identified from the data

A second is classified as closed if it sits inside a run of at least ten minutes in which none of the three pairs printed a new price. Ten minutes of simultaneous silence across three majors does not occur while the market is open. Eight closures are found, all 48 hours to within two minutes, and no trading calendar is consulted. About 3.86 million open-market seconds remain.

2.4 Two measurement decisions

The clock

Vendor timestamps are documented as EST, but the sample lies entirely inside US daylight saving, so the operative offset is UTC−4. The Friday-to-Sunday week boundary does not establish this, since FX closes at 17:00 New York local year-round and that boundary appears under either offset. The offset was determined by scanning candidate offsets against UTC-stamped ticks from a second vendor over a control window. At −4h the correlation is 1.00000 with a gap standard deviation of 0.01 pips; at −5h it is −0.71 and 48.65 pips.

Quote pairing

Bid and ask are carried through the grid beside the mid and are always taken from a single tick. Three ARG_MAX calls over one ordering key resolve to the same winning row, so the bid and ask columns are null in identical rows and the forward fill cannot pair a bid from one second with an ask from another. Taking max(ask) with min(bid) across a second would produce a tighter aggregate and a spread no participant was shown. Section 6 depends on this pairing, so it is asserted after the splice rather than assumed.

3 Exact Bayesian segmentation

3.1 The feature

For each retained hour h , dispersion is summarised by the median absolute deviation rescaled to standard-deviation units:

$$\text{MAD}_h = 1.4826 \cdot \text{median}_{t \in h} |z_t - \text{median}_{s \in h} z_s|.$$

The basis has excess kurtosis of 277 over the full sample, so the hourly standard deviation is determined by a small number of seconds while the MAD is not. The model is fitted to $\log \text{MAD}_h$, which makes a shift multiplicative and makes the feature close to symmetric over the roughly 3600 observations per hour.

Two nuisance profiles are subtracted. With $\text{slot}(h)$ the within-day position of the bucket and $\text{age}(h)$ the buckets elapsed since the session reopened,

$$y_h = \log \text{MAD}_h - m(\text{slot}(h)) - r(\min(\text{age}(h), R)), \quad (3)$$

where m and r are medians taken over the full sample. The diurnal amplitude is roughly fourfold peak to trough. The reopen term is required because a weekend closure is followed by a thin Sunday evening, so the first buckets of each week sit systematically high; without it, a regime ending during a closure is recorded a few hours after the reopen, since no bucket exists inside the gap for a changepoint to occupy.

Both profiles are estimated over the whole sample and therefore cannot have been fitted either side of a candidate date. A common profile cannot create a level shift, only reduce one, so the adjustment works against the finding.

3.2 Marginal likelihood of a segment

Write a segment as a one-regressor regression

$$\zeta_t = \mu \omega_t + e_t, \quad e_t \sim \mathcal{N}(0, \sigma^2),$$

which reduces to the ordinary segment mean when $\omega_t \equiv 1$ and also accommodates the AR(1) transform below without modification. Place a Normal-Inverse-Gamma prior,

$$\mu \mid \sigma^2 \sim \mathcal{N}\left(m_0, \frac{\sigma^2}{\kappa_0}\right), \quad \sigma^2 \sim \text{Inv-Gamma}(a_0, b_0),$$

and let n , $S_{\omega\omega} = \sum \omega_t^2$, $S_{\omega\zeta} = \sum \omega_t \zeta_t$ and $S_{\zeta\zeta} = \sum \zeta_t^2$ be the sufficient statistics. Integrating out (μ, σ^2) gives a closed form. With

$$\kappa_n = \kappa_0 + S_{\omega\omega}, \quad a_n = a_0 + \frac{n}{2}, \quad m_n = \frac{\kappa_0 m_0 + S_{\omega\zeta}}{\kappa_n}, \quad b_n = b_0 + \frac{1}{2} (S_{\zeta\zeta} + \kappa_0 m_0^2 - \kappa_n m_n^2),$$

the log marginal likelihood is

$$\log p(\text{segment}) = \log \Gamma(a_n) - \log \Gamma(a_0) + a_0 \log b_0 - a_n \log b_n + \frac{1}{2} (\log \kappa_0 - \log \kappa_n) - \frac{n}{2} \log 2\pi. \quad (4)$$

Every input to (4) is available in $O(1)$ from cumulative sums, which is what makes exhaustive enumeration cheaper than sampling.

The prior is weak but proper. $\kappa_0 = 0.01$ is one hundredth of an hour's worth of information about the segment mean, and $a_0 = 2$ is the smallest shape giving a finite prior mean for σ^2 . Propriety matters because improper priors make the model comparison below undefined. The scale b_0 is set from the dispersion of the series being fitted, which is empirical Bayes; that dispersion is inflated by the shift under test, so the choice reduces the evidence for a change.

3.3 AR(1) errors across a market that closes

Hourly dispersion is strongly autocorrelated because volatility clusters. Under independent errors a run of above-average hours is not distinguishable from a level shift, and the estimator then selects an excursion on series containing no changepoint. Errors within a segment are therefore modelled as AR(1) with coefficient ρ .

Whitening requires two cases, because the market closes weekly. Where bucket h is adjacent in clock time to $h - 1$, the one-step-ahead form applies:

$$\zeta_h = y_h - \rho y_{h-1}, \quad \omega_h = 1 - \rho.$$

Where it is not, meaning the first bucket of the sample and the first bucket after every closure, there is no predecessor and the stationary marginal is used:

$$\zeta_h = y_h \sqrt{1 - \rho^2}, \quad \omega_h = \sqrt{1 - \rho^2}.$$

Both place the observation on the same σ , so (4) applies unchanged. The stationary form is also used at the first bucket of every segment, whose predecessor belongs to a different segment with a different mean.

ρ is marginalised over a grid $\mathcal{R}$ spanning $[0, 0.90]$ under a uniform prior rather than profiled, so the reported evidence accounts for not knowing it:

$$\log Z_M = \log \sum_{\rho \in \mathcal{R}} \exp(\log Z_M(\rho)) - \log |\mathcal{R}|.$$

The grid includes zero, so the independent-error model is nested and the data can return it. The upper end stops short of one, where a level shift and a random walk are not separately identified.

3.4 Model space and enumeration

Four models are compared over n retained buckets, with m the minimum admissible segment length:

M_0	no change	one segment $[0, n)$
M_1	permanent shift	one changepoint τ
M_2	excursion	two changepoints; the outer segments share one set of parameters
M_3	two free changes	two changepoints; three free segments

M_0 is included because a one-changepoint model always returns a changepoint. Without a no-change alternative in the comparison, the selected date carries no information about whether a change occurred. M_2 separates the two substantive hypotheses: a new regime against a bounded window followed by a return.

Changepoints are discrete and there are at most two, so the posterior is computed by enumerating every admissible configuration. Writing $\mathcal{T} = \{(\tau_1, \tau_2) : \tau_1 \geq m, \tau_2 - \tau_1 \geq m, n - \tau_2 \geq m\}$ and ℓ for the sum of (4) over the segments a configuration induces, the evidence under a uniform prior over configurations is

$$\log Z_{M_k} = \log \sum_{c \in \mathcal{C}_k} \exp(\ell(c)) - \log |\mathcal{C}_k|. \quad (5)$$

The result is exact. There are no chains, no convergence diagnostics, and the output is identical on every run. Posterior model probabilities follow from (5) with equal model priors.

M_2 and M_3 share their three blocks. M_2 pools the outer two into one regime, which is exact because sufficient statistics are additive across conditionally independent segments, while M_3 gives them separate means. Computing the three blocks once and combining them reduces the work from five passes over the configuration set to three.

Marginal posteriors over each changepoint follow by summation. The excursion indicator is obtained from a difference array, so the cost is $O(|\mathcal{T}|)$ rather than $O(|\mathcal{T}| \cdot n)$:

$$p_{\text{onset}}(t) = \sum_{(t, \tau_2) \in \mathcal{T}} W(t, \tau_2), \quad \Pr(h \text{ elevated}) = \sum_{\tau_1 \leq h < \tau_2} W(\tau_1, \tau_2), \quad W \propto e^\ell.$$

Reported intervals are the shortest contiguous window carrying 95% of the marginal rather than a highest-density set. The posterior over a changepoint is often multimodal, and a set of disjoint hours does not correspond to a date range. The interval is conservative where modes separate.

3.5 Result and calibration

Model	Changepoints	$\log Z$	Posterior
M_0 no change	0	605.8	2.7×10^{-43}
M_1 permanent shift	1	645.4	4.5×10^{-26}
M_2 excursion	2	702.8	0.384
M_3 two free changes	2	703.3	0.616

Table 1: Marginal likelihood of each segmentation of hourly log dispersion. Posterior mass on the two-changepoint models is 1.000 to three decimals. Source: 02_model_evidence.csv.

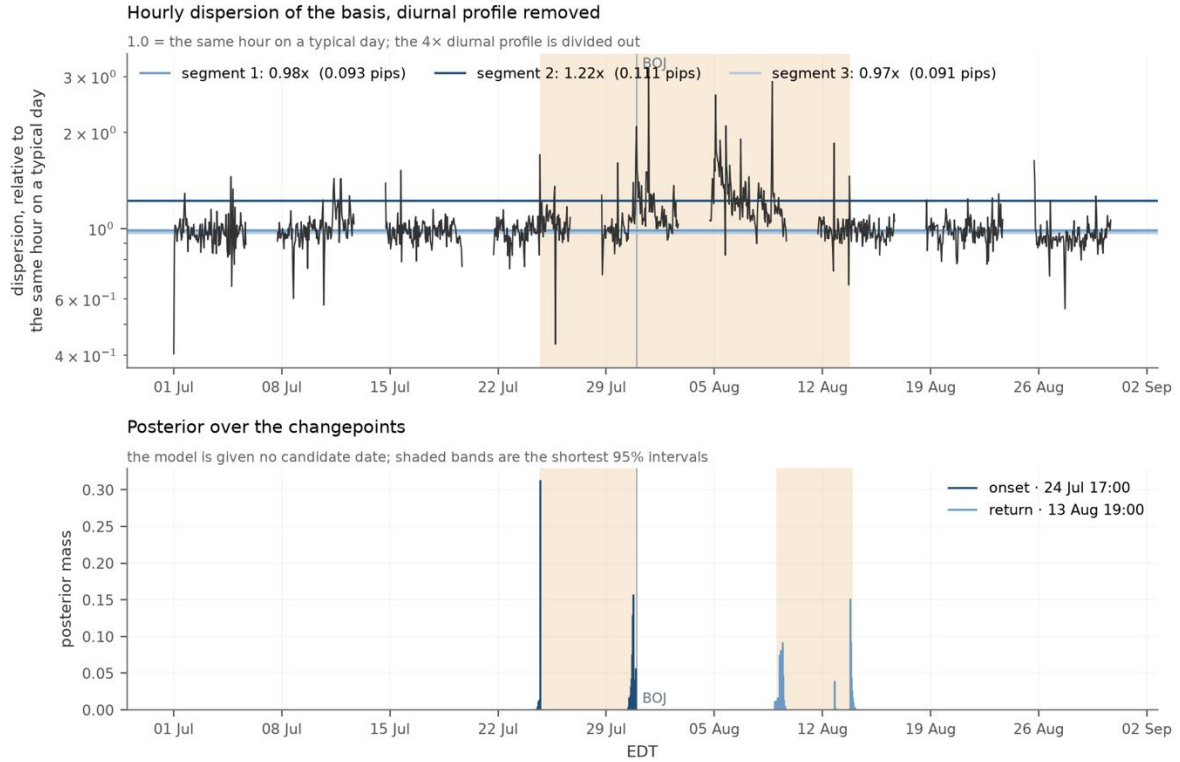


Figure 1: Hourly dispersion with the diurnal profile divided out, and the marginal posterior over each changepoint. The onset posterior is bimodal, with mass at 24 July and again at the policy decision on 30 July; the return is better localised. Fitted segment levels are 0.093, 0.111 and 0.091 pips, a within-hour ratio of 1.19 to 1.22 against a pooled ratio of 3.09 discussed in Section 7. No candidate date was supplied to any model.

A Bayes factor measures how much better M_2 fits than M_0 on this series. It does not establish that a series without a changepoint would have looked different. One hundred replicates were simulated with the sample's length, session structure, fitted ρ and residual scale, and no change anywhere, then refitted with the same code.

Pr(any change selected no change)	0.020
Pr(excursion selected no change)	0.010
Null log BF, 95th percentile	-2.8
Observed log BF (M_2 against M_0)	+97.1
Fraction of null replicates the observation exceeds	1.000

The AR(1) term produces that calibration. Under independent errors the same code selects an excursion on noise in roughly four series out of five. All twenty specifications in `02_sensitivity.csv` select a two-changepoint model, varying the summary statistic, bucket width, prior scale, minimum segment length, both admission filters, the deseasonalisation, the rollover boundary and the error model. None selects M_0 or M_1 .

4 Error correction under a known cointegrating vector

4.1 Specification

With the cointegrating vector imposed, the system is a vector error-correction model:

$$\Delta x_t = c + \alpha \tilde{z}_{t-1} + \sum_{i=1}^p \Gamma_i \Delta x_{t-i} + e_t, \quad \alpha \in \mathbb{R}^3, \Gamma_i \in \mathbb{R}^{3 \times 3}, \quad (6)$$

fitted separately inside each estimated regime. A VAR in returns would discard $\tilde{z}_t$, which is the only variable in the system that is not a martingale and the only one the parity condition constrains.

The lag order is $p = 21$ seconds, the cap of the tested grid $\{1, 2, 3, 5, 8, 13, 21\}$. BIC is still improving at the cap, which is recorded rather than presented as clean selection; `03_sensitivity.csv` refits at $p \in \{1, 5\}$ and the ordering of the regimes is unchanged.

4.2 $\lambda = w^\top \alpha$ is an identity

From (2), $\tilde{z}_t = w^\top x_t = w^\top (x_{t-1} + \Delta x_t)$, so

$$\tilde{z}_t = \tilde{z}_{t-1} + w^\top \Delta x_t \quad (7)$$

holds exactly. Pre-multiplying (6) by $w^\top$ and substituting,

$$\Delta \tilde{z}_t = w^\top c + \underbrace{(w^\top \alpha)}_{\lambda} \tilde{z}_{t-1} + \sum_{i=1}^p (w^\top \Gamma_i) \Delta x_{t-i} + w^\top e_t.$$

λ is therefore the one-step closure rate of the basis, obtained from the system without a separate regression.

4.3 The sign restriction is a test

Signs are predicted before estimation. If $\tilde{z} > 0$ the direct rate is rich against the synthetic, so closure requires the direct leg to fall and the synthetic to rise:

$$\alpha_{AJ} < 0, \quad \alpha_{AU} > 0, \quad \alpha_{UJ} > 0, \quad \text{hence} \quad \lambda = -\alpha_{AU} - \alpha_{UJ} + \alpha_{AJ} < 0.$$

All four hold in all three regimes at $|t|$ between 2.4 and 11.5 (`04_weak_exogeneity.csv`).

4.4 Two half-lives

λ describes the first second only. The transitory terms can accelerate or delay the remainder, so the decay of a full dislocation requires the system. Define the state

$$s_t = (\tilde{z}_t, \Delta x_t^\top, \Delta x_{t-1}^\top, \dots, \Delta x_{t-p+1}^\top)^\top \in \mathbb{R}^{1+3p}.$$

Combining (6) with the identity (7) gives $s_t = Ms_{t-1}$ in the absence of shocks, where

$$M = \begin{pmatrix} 1 + w^\top \alpha & w^\top \Gamma_1 & \cdots & w^\top \Gamma_p \\ \alpha & \Gamma_1 & \cdots & \Gamma_p \\ 0 & I_3 & & \\ & & \ddots & \\ 0 & & I_3 & 0 \end{pmatrix}. \quad (8)$$

The first row of (8) is the identity (7), not an additional assumption. The decay path of a unit dislocation is $h(k) = (M^k e_1)_1$, and the full-system half-life is the first crossing of $\frac{1}{2}$, interpolated between the bracketing steps.

The spectral radius of M is the stationarity check the error-correction reading requires. With the cointegrating vector imposed the system carries no unit root in this state, so an eigenvalue at or above one would mean the regime's basis is not mean-reverting. Observed values are 0.9889, 0.9957 and 0.9841.

4.5 Precision is not uniform across α

The three components of α are estimated far less precisely than their combination. Each leg's own volatility is orders of magnitude larger than the basis, so regressing that leg's return on the basis carries a large error variance, while λ is a combination in which the common moves cancel by construction. The closure-rate claim is consequently better supported than the leg-split claim, and the tables do not present them as equally reliable.

4.6 HAC for a single linear functional

Increments carry first-order autocorrelation between -0.30 and -0.42 from bid-ask bounce, so OLS standard errors are wrong by a non-negligible factor. Standard errors are Newey-West.

The full HAC matrix for a three-equation system with K regressors is $3K \times 3K$ and requires lagged cross-products of an $(n, 3K)$ array. At a million seconds and sixty-five regressors it is not worth forming, and it is not needed. Only the $\tilde{z}_{t-1}$ coefficient is of interest, and for a single linear functional the sandwich collapses. Writing $u_t \in \mathbb{R}^3$ for the residual vector and e_j for the selector of the error-correction column,

$$\hat{\alpha} - \alpha = \sum_t \underbrace{e_j^\top (X^\top X)^{-1} x_t}_{h_t \in \mathbb{R}} u_t^\top,$$

so the required 3×3 covariance is the long-run covariance of the single vector series $\varphi_t = h_t u_t$.

That covariance is computed from overlapping block sums rather than a loop over lags. Let $B_s = \sum_{i=0}^{m-1} \varphi_{s+i}$. Expanding,

$$\frac{1}{m} \sum_s B_s B_s^\top = \frac{1}{m} \sum_s \sum_{i,j=0}^{m-1} \varphi_{s+i} \varphi_{s+j}^\top = \sum_{|k| < m} \frac{m - |k|}{m} \Gamma_k = \sum_{|k| < m} \left(1 - \frac{|k|}{m}\right) \Gamma_k,$$

since exactly $m - |k|$ ordered pairs (i, j) in $[0, m)^2$ satisfy $i - j = k$. This reproduces the Bartlett kernel with truncation $m - 1$ exactly, up to the first and last $m - 1$ rows of each contiguous run.

With runs hours long and m of the order of a minute, that edge term is around 10^{-5} of the total. Blocks that would straddle a break are refused.

4.7 Row admission

A lag window is admitted only where the whole span of $p + 2$ consecutive seconds is contiguous in clock time, outside the daily rollover window, inside one regime and one trading day, and labelled with regime posterior at least 90%. Nothing is differenced across a weekend, a closure, a rollover exclusion or a regime boundary.

Using the posterior rather than the hard boundary places a buffer around each changepoint that is derived from the data instead of chosen. Both changepoints span market closures, so some buffer is required. The row set is fixed once at the lag cap and reused for every lag order and every regime, so no two numbers in the output are computed on different samples.

The segment boundaries come from the median segmentation of Section 3 and are read from the labels rather than typed as dates. The onset boundary is a choice between two posterior modes and is discussed in Section 8.2; the alternative is carried as a row of `03_sensitivity.csv` so its cost can be read off.

One further choice matters for Section 7. The error-correction term is centred on the day. Centring on the regime leaves drift in the daily mean inside the residual, where the estimator reads it as persistence: that variant returns a stress half-life of 967 seconds against a baseline of 155.

5 Attribution

Three routes, sharing no machinery, in increasing order of what each assumes.

5.1 By identity

From (7), the change in the basis over any window is exactly the sum of three leg contributions:

$$\tilde{z}_{t_1} - \tilde{z}_{t_0} = \sum_{k \in \{AU, UJ, AJ\}} w_k (x_{t_1}^k - x_{t_0}^k).$$

Applying this to the opening and closing window of each detected episode splits both by arithmetic, with no model and no lag order.

Share of the move	pre	stress	post
Opening: AUD/USD + USD/JPY	0.86	1.12	1.05
Closing: AUD/JPY	0.64	0.71	0.48
Median seconds to open	2	19	4
Median seconds to close	3	24	4

Table 2: Episode-level attribution. Shares can exceed one because leg contributions can oppose. Source: 04_attribution.csv.

5.2 By correlation and by test

Cross-correlations between each leg's increments and the basis's are computed at matched leads and lags out to a minute, on pairs inside one unbroken run. Block Granger tests are then run inside (6).

A degeneracy has to be handled. The basis is a fixed combination of the three legs, so at every lag the three legs' increments together span the basis's own lagged increment. Dropping any one leg's block removes part of the basis's own autoregression along with that leg's information, and when the basis is persistent all three blocks then test large for that reason alone. The headline test therefore conditions on the basis's own history explicitly: for each leg, a smaller model containing the basis level, the basis's own lagged increments and that one leg's lagged increments, asking what the leg adds beyond the basis's own memory. The full-system figure is reported beside it as a control. Neither model is refitted; both are column subsets of one extended design whose cross-products are accumulated once.

5.3 Effect sizes rather than p -values

Every test runs on between one hundred thousand and a million seconds. At that size a coefficient of no economic consequence rejects at any conventional level. Each test therefore carries an effect size, the incremental predictable move in pips per second, beside a placebo threshold computed from the same statistic with that leg's increments drawn from a different

day. The placebo preserves the leg's own autocorrelation and volatility clustering and destroys only the cross-leg timing, so it is the distribution of the statistic when the null holds and everything else is unchanged.

The results separate cleanly. Cross-leg effects clear their placebo by 3.5 to 10.6 times. Leg-to-basis effects, conditioned on the basis's own memory, explain between 0.01% and 0.07% of basis increment variance; AUD/JPY fails Holm correction in all three regimes and USD/JPY in two. The basis is close to unforecastable from its own components while those components remain highly forecastable from each other.

5.4 Why no Hasbrouck or Gonzalo-Granger shares

Those measures are defined for two prices of one asset: one cointegrating vector, one common trend, and loadings that split into shares summing to one. This system has three prices and one restriction, so it has two common trends, an AUD trend and a JPY trend, and $\alpha_{\perp}$ is 3×2 rather than a vector. No scalar share per leg is identified without a normalisation the data does not supply. The weak-exogeneity tests are the part of that framework this system supports, and they are what is reported.

6 The transaction-cost band

6.1 The two round trips

Sections 3 and 4 work on mid quotes, which is the correct series for measuring how prices adjust and the wrong one for asking whether a dislocation was executable. A round trip crosses three spreads. Writing b and a for bid and ask in AUD/JPY price units,

$$e_A = AJ_b - AU_a UJ_a \quad (\text{sell direct, buy synthetic}), \quad e_B = AU_b UJ_b - AJ_a \quad (\text{buy direct, sell synthetic}).$$

Each is the profit per unit of AUD from one complete circuit at prices that were quoted. They cannot be positive together, since $e_A > 0$ and $e_B > 0$ would require $AJ_b > AJ_a$.

6.2 The band, and the exact cancellation

Let z be the mid basis. Then $e_A = z - c_A$ and $e_B = -z - c_B$ with

$$c_A = (AJ_m - AJ_b) + (AU_a UJ_a - AU_m UJ_m), \quad c_B = (AJ_a - AJ_m) + (AU_m UJ_m - AU_b UJ_b),$$

so nothing is executable while $z \in [-c_B, +c_A]$. The width of that band is

$$W = c_A + c_B = \underbrace{(AJ_a - AJ_b)}_{s_{AJ}} + (AU_a UJ_a - AU_b UJ_b).$$

Writing each quote as its mid plus or minus half a spread, the second term expands as

$$(AU_m + \frac{s_{AU}}{2})(UJ_m + \frac{s_{UJ}}{2}) - (AU_m - \frac{s_{AU}}{2})(UJ_m - \frac{s_{UJ}}{2}) = AU_m s_{UJ} + UJ_m s_{AU},$$

the quadratic terms cancelling in the difference. Therefore

$$\boxed{W = s_{AJ} + AU_m s_{UJ} + UJ_m s_{AU}} \quad (9)$$

holds exactly rather than to first order. The band decomposes additively across the three legs with no interaction term, so which leg sets the bound is determined by arithmetic. Nothing in (9) is estimated and no cost parameter is assumed.

6.3 The band in units of the basis

Section 3 measured a numerator. Equation (9) supplies the denominator. The quantity of interest is $W / \text{MAD}(z)$ per regime, computed on the same seconds.

Both quantities widened. The dispersion grew by 2.35 and the band by 1.29, so the band-to-gap ratio approximately halved. The basis moved closer to the band and remained an order of magnitude inside it.

	pre	stress	post
Band width W , median (pips)	2.61	3.35	2.57
Basis dispersion, MAD (pips)	0.112	0.264	0.095
Band / dispersion	23.3	12.7	27.1

Table 3: Rows restricted to seconds where all three legs quoted within one second and the regime posterior is at least 90%. Source: 05_execution.csv.

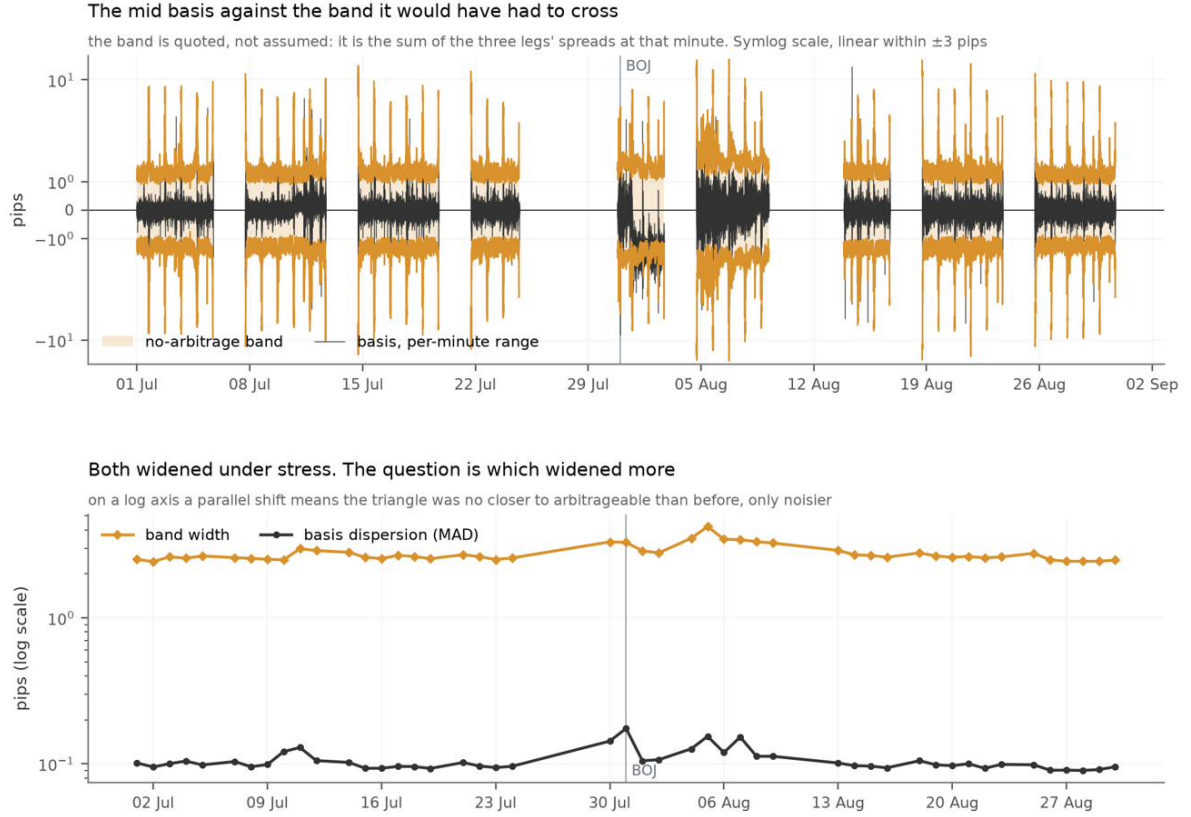


Figure 2: The mid basis against the band it would have had to cross. On the log axis of the lower panel a parallel shift indicates that the band-to-gap ratio did not change.

6.4 Sensitivity to the spread level

Both vendors publish aggregator quotes, which are wider than interdealer, so the comparison is biased toward finding no executable dislocation. Every spread is therefore scaled by a factor k , equivalent to $e_A(k) = z - kc_A$ and $e_B(k) = -z - kc_B$ with the mid basis held fixed, and the share of seconds with a positive edge is reported as a function of k . Halving every quoted spread leaves 99.95% of calm seconds inside the band; only at a tenth of quoted do the calm regimes reach 22.6% and 14.4% (05_haircut.csv).

6.5 Why a positive edge is not an opportunity

The grid is one second wide and the three legs are separate feeds, so a row records the last print of each leg inside that second rather than three prices known to have stood together. Triangular parity requires simultaneity across all three. A positive edge computed from this grid is therefore an upper bound on what was available and not a measurement of it: when prices travel far enough inside a second, the separation between prints produces an apparent edge, and second resolution cannot distinguish the two readings. No finer data exists for these pairs from either vendor.

Two diagnostics are reported in place of an assertion. A real opportunity decays as the delay between signal and fill grows and a measurement artefact does not; the observed count is flat across execution delays of zero to five seconds. A displacement that was present is indifferent to how fast the market was moving and non-simultaneity is not; the count is not concentrated in the fastest quintiles of within-second travel. Both point away from reading the stress count as opportunity.

7 Reconciliation

Five routes to the closure speed of the basis are distributed across the preceding sections and they do not agree.

Route	What it estimates	Stress / calmest
Episode ruler	time for a large excursion to halve	9.3
AR(1) on the level	decay of an average deviation, one parameter	9.7
VECM, one-step	decay implied by λ alone	2.9
VECM, full system	decay of the 21-lag system run down	4.1
VECM, day by day, median	the typical day rather than the typical second	1.4

Table 4: Source: 06_closure_reconciliation.csv.

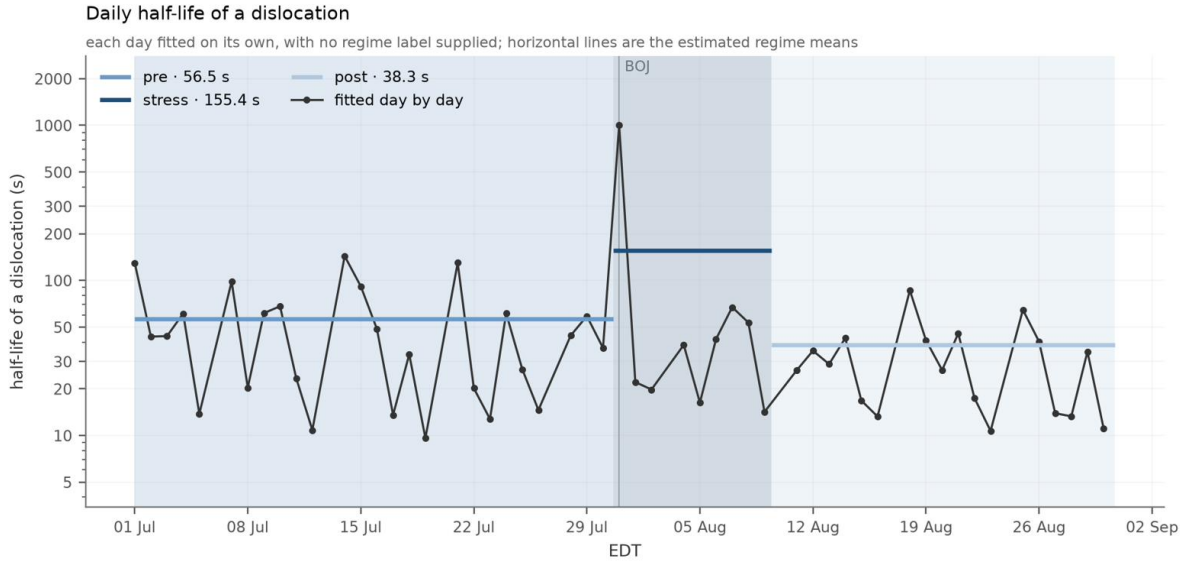


Figure 3: Each day fitted on its own, with no regime label supplied. The pooled stress mean of 155 seconds sits above every stress day but one, which is the disagreement the rest of this section accounts for. Discussed in Section 7.4.

7.1 Decomposing the level change

For a stationary AR(1) level with innovation scale σ and coefficient ρ ,

$$\text{sd}(z) = \frac{\sigma}{\sqrt{1-\rho^2}}, \quad \text{so} \quad \log \frac{\text{sd}(z)_A}{\text{sd}(z)_B} = \underbrace{\log \frac{\sigma_A}{\sigma_B}}_{\text{noise}} + \underbrace{\log \frac{\sqrt{1-\rho_B^2}}{\sqrt{1-\rho_A^2}}}_{\text{persistence}}. \quad (10)$$

The split is additive on the log scale. The innovation scale of the basis does not need separate estimation, because $\varepsilon_z = w^\top e$ and therefore

$$\sigma^2 = w^\top \Sigma w = \sigma_{AU}^2 + \sigma_{UJ}^2 + \sigma_{AJ}^2 + 2\sigma_{AU}\sigma_{UJ}\rho_{AUUJ} - 2\sigma_{AU}\sigma_{AJ}\rho_{AU AJ} - 2\sigma_{UJ}\sigma_{AJ}\rho_{UJ AJ}, \quad (11)$$

reconstructible from the per-leg residual scales and correlations that (6) already produces. Substituting them reproduces the measured level dispersion to within 6% to 17% across all three regimes, which confirms that (10) and (11) describe the same object.

7.2 What the pooled statistic measures

Equation (10) attributes the widening to larger shocks and slower reversion. That attribution is incomplete, because $\text{sd}(z)$ is a pooled quantity. Splitting the level variance on the trading day d ,

$$\text{Var}(z) = \underbrace{\mathbb{E}_d[\text{Var}(z \mid d)]}_{\text{within-day}} + \underbrace{\text{Var}_d(\mathbb{E}[z \mid d])}_{\text{day-to-day}}. \quad (12)$$

Both terms are evaluated on the same seconds by `07_boundary_sweep.py`, which accumulates per-hour sums once and aggregates them into per-day cells. The same widening then reads differently at four nested scopes, and nothing about the market changes between the rows.

Scope of the measurement	Stress / calm
Within the hour (MAD)	1.21
Within the day (sd)	1.98
Pooled across the window, robust (MAD)	2.91
Pooled across the window, non-robust (sd)	4.22
Day-to-day share of level variance, stress	80.6%
Day-to-day share of level variance, calm	11.6%

Table 5: Applying (12) at the segmentation in use. Calm pools the pre and post segments. Source: `07_boundary_named.csv`.

Four fifths of what the pooled statistic calls dispersion under stress is the level moving between days. Each step up the ladder admits another layer of that movement. The within-hour figure of 1.21 agrees with the 1.19 to 1.22 the changepoint model returned independently from hourly buckets, which is the check that the decomposition and the fit describe the same object.

7.3 The decomposition does not turn on the regime boundary

Section 8.2 records that the onset posterior is bimodal and that the boundary in use is the median rather than the mode. Every quantity above is conditioned on that choice.

Rather than evaluate the decomposition at one boundary, or at two, `07_boundary_sweep.py` evaluates it at every hourly onset between 23 and 31 July under both candidate returns, 128 boundaries in total. The decomposition is a closed form in per-day sufficient statistics, so a boundary costs $O(\text{hours})$ rather than $O(\text{seconds})$. Nothing is refitted; the basis series is re-aggregated.

	median rule (in use)	modal rule	all 128 boundaries
Onset	30 Jul 15:00	24 Jul 17:00	23 to 31 Jul
Return	09 Aug 10:00	13 Aug 19:00	both
Stress window (hours)	183	331	160 to 354
Day-to-day share, stress	80.6%	78.7%	78.7% to 87.3%
Day-to-day share, calm	11.6%	11.8%	10.2% to 12.9%
Within-hour MAD ratio	1.21	1.13	1.12 to 1.21
Pooled MAD ratio	2.91	1.63	

Table 6: The decomposition as a function of the boundary. Source: 07_boundary_named.csv and 07_boundary_sweep.csv.

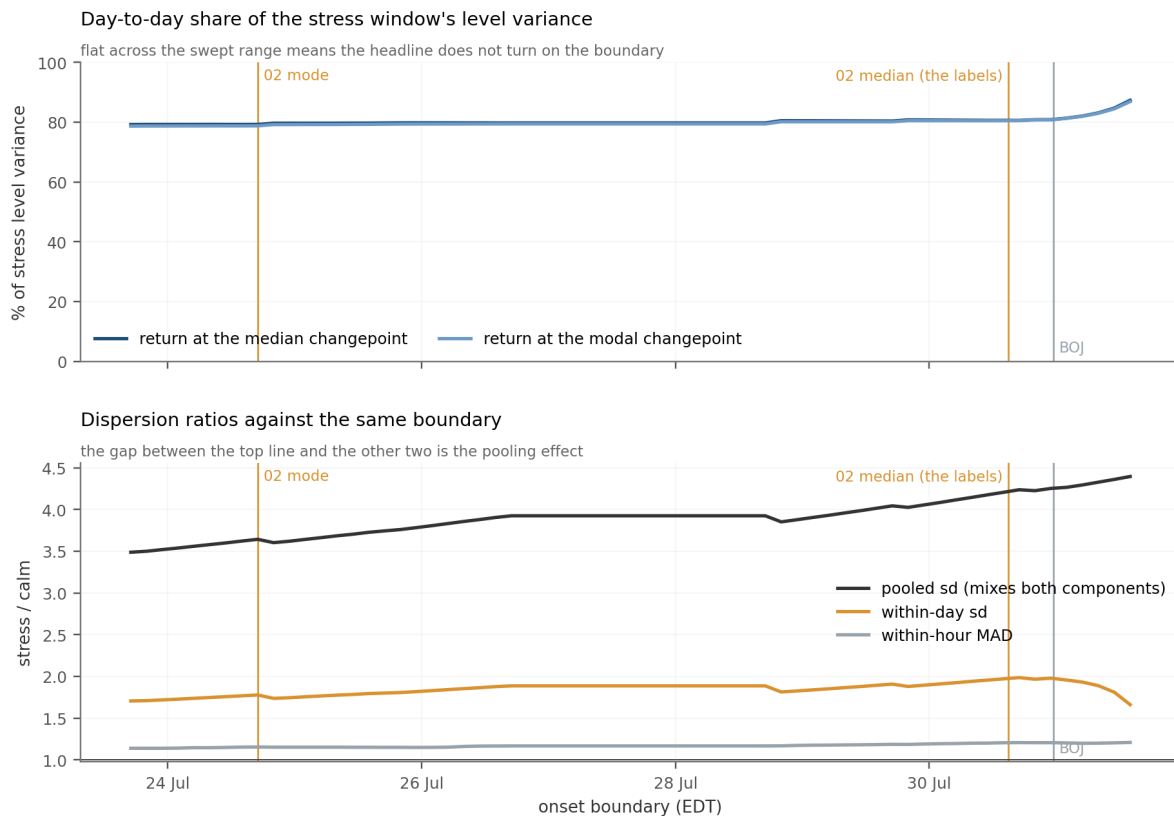


Figure 4: The decomposition at every candidate onset, under both candidate returns. The upper panel is flat across the swept range, so the headline does not turn on the boundary. In the lower panel the pooled ratio rises with the boundary while the two decomposed ratios do not, and the gap between them is the pooling effect.

The day-to-day share stays between 78.7% and 87.3% across every boundary tested, against 10.2% to 12.9% in the calm segments, over stress windows ranging from 160 to 354 hours. The within-hour widening moves by 8% across the whole sweep, which makes it the most stable quantity in the project.

The pooled MAD ratio is the exception, falling from 2.91 to 1.63 when the boundary moves to the modal onset. The quantity this report treats as the headline is stable under the boundary choice; the quantity it argues against reporting is the one that is not.

7.4 Effect on the persistence estimates

Write $z_t = \mu_d + u_t$ with μ_d constant within a day and u_t an AR(1) with coefficient ρ_w . Almost every lag-one pair lies inside a day, so

$$\rho_{\text{pooled}} = \frac{\sigma_{\text{day}}^2 + \rho_w \sigma_{\text{within}}^2}{\sigma_{\text{day}}^2 + \sigma_{\text{within}}^2}.$$

Inverting at the shares above:

	pre	stress	post
Pooled AR(1), as reported	0.851	0.976	0.792
Implied within-day AR(1)	0.825	0.723	0.786
Implied within-day half-life (s)	3.6	2.1	2.9

The stress regime has the lowest within-day persistence of the three. Its pooled AR(1) of 0.976 is almost entirely the daily-mean component.

This is a two-component calculation on published values rather than a re-estimation. It is corroborated by a route sharing none of its machinery: the VECM refitted day by day removes the daily level by construction and returns median half-lives of 43.6, 38.4 and 27.6 seconds, placing the stress regime between the other two.

Three estimators that remove the daily level return a small effect. Three that retain it return a large one. The choice of estimand moves the reported figure by a factor of seven.

7.5 One day, not one regime

The daily fits in Figure 3 resolve what remains. Eight of the nine stress days fit at 14 and 67 seconds, inside the pre-stress range of 10 to 144. The ninth is 31 July, the session following the announcement at 30 July 23:30 in the data's clock, at 1,003 seconds: seven times slower than the next slowest day in the sample and fifteen times slower than the next slowest stress day. The pooled stress mean of 155 seconds is above every stress day except that one.

Across the conventions in `03_sensitivity.csv` the ratio spans 1.5 to 24.8. The direction is stable across all of them; the magnitude is not determined by the data alone. The supportable statement is that one day was extraordinary.

8 Limitations

8.1 Two days fail the data-quality gate

The daily-bias check of Section 2 rejects seven of 53 days at a threshold of 0.10 pips. Two of them by a factor of fifteen.

Day	AUD/USD	USD/JPY	AUD/JPY	Open-market seconds
11 Jul	+0.143	+0.338	−0.228	86,400
12 Jul	+0.137	+0.320	−0.217	61,200
31 Jul	−0.045	−0.104	+0.068	86,400
1 Aug	−0.977	−2.247	+1.464	86,400
2 Aug	−0.991	−2.252	+1.468	61,200
8 Aug	+0.176	+0.393	−0.258	86,400
9 Aug	+0.180	+0.402	−0.265	61,200

Table 7: Median signed gap between each leg and the value the other two imply, in that leg’s own pips. Source: 00_daily_bias.csv.

The gate returns a non-zero exit code on this sample. It exists to catch a persistent sub-pip bias, which the tail statistic cannot see: the 99.9th percentile of the absolute gap passes at 3.21, 7.29 and 4.83 pips against a 25-pip threshold.

The triangle cannot attribute the discrepancy to a leg. The three columns are one discrepancy expressed three ways and any single leg reproduces it exactly. Attribution requires a reference outside the triangle, and none is available: Dukascopy carries the same level on those days, but HistData sources from Dukascopy, so agreement between them establishes that the level is present in a shared upstream feed rather than that it is real. Settling it requires a feed from a different liquidity pool. The offset is documented and carried, not fixed.

1 and 2 August carry a standing basis offset near -1.47 pips across 147,600 open-market seconds, which is 22% to 24% of the stress regime depending on the admission rule. The implied contribution to the regime mean basis is -0.33 to -0.35 pips. The measured stress mean basis is -0.27 pips, against $+0.016$ pre and $+0.005$ post. The two days account for the entire mean level shift attributed to the stress regime.

Estimates insensitive to this: the changepoint dates, because a level offset constant within a day leaves within-hour dispersion unchanged and neither estimated changepoint falls near 1 or 2 August; and the error-correction coefficients, because the VECM centres on the day.

Estimates sensitive to it: the regime-level standard deviation, the pooled AR(1) of the level, and the positive-edge counts of Section 6.

8.2 The onset is bimodal, and the regime boundary is a choice

The onset posterior has two modes. One sits at 24 July 17:00, the other at the policy decision on 30 July, and the 95% window between them spans a weekend closure. Across the 17 same-feature specifications of 02_sensitivity.csv the split is 11 at 24 July, 4 at 30 July, 1 at 29 July and 1 at 1 July.

The boundary this project uses is 30 July, which is not the date the estimator selects most often. There are two reasons and they are of different kinds.

The first is methodological. The per-second labels written by `02_changepoint_detection.py` are the *median* segmentation: a bucket is called elevated when more than half the posterior mass says it is. 30 July is what that rule returns. 24 July is the posterior mode, a taller single spike carrying less than half the total mass. Reporting the mode while labelling on the median would leave the headline describing a different window from the one every downstream script consumes.

The second is historical, and was a researcher degree of freedom rather than a result. The project was originally scoped around the 31 July policy decision, so the boundary sitting 8.5 hours before that announcement is the one it was built to find. That is a prior, and Section 7.3 replaces it with a measurement: the decomposition is evaluated at every candidate boundary rather than at the chosen one.

The cost of the alternative is measured rather than argued. Taking the 24 July onset moves 338,396 seconds from pre into stress and the closure ratio falls from 4.06 to 3.35 (`03_sensitivity.csv`, row 02's *modal onset*, 24 Jul 17:00). The chosen boundary is therefore the one under which the estimated shift is larger, and a reader who prefers the modal onset should read 3.35 and adjust Section 7 accordingly.

One observation bears on which mode is the more credible without settling it. The 24 July 17:00 bucket covers 17:30 to 17:59:59, and it contains two of the twenty widest episodes in the sample: 7.335 pips at 17:44:08, the widest in the entire pre-stress segment, and 5.65 pips at 17:58:14. Both fall in the half hour after the rollover filter reopens, which the boundary audit of the next subsection shows carries 21.5 times the expected episode rate. Part of the 24 July mode may therefore be the filter boundary rather than the market. That is not resolved here.

A change falling inside a closure can only be recorded at the reopen, so the tables report the bracket the data supports rather than a modal hour. The return is better behaved, clustering at 12 to 13 August in 14 of the 17 same-feature specifications.

8.3 Censoring at the filter boundary

Episode detection excludes 17:00 to 17:30, so the detector sees a session opening at 17:30:00 and closing at 16:59:59. Nine of the twenty widest episodes start in hour 17, which holds 2.1% of admissible seconds, at 21.5 times the expected rate. Two begin exactly at 17:30:00 and three end exactly at 16:59:59, against an expectation of 0.0002 for each. These are censored observations rather than starts and ends, and the widest episode in the sample, 8.911 pips on 4 August, is one of them. Moving the boundary to 18:00 leaves the selected segmentation unchanged.

8.4 Scope

One event, two months, one currency triangle. The results describe a single episode and are not distributional claims about FX stress in general. The quotes are retail aggregator quotes, which biases Section 6 toward finding no executable dislocation; the haircut ladder quantifies that bias rather than removing it.

8.5 Summary

Supported by the evidence:

- Hourly dispersion is best described by a bounded excursion that returns rather than a permanent shift.
- Within-hour dispersion rose by 1.21 times, and that figure is stable across all 128 candidate regime boundaries.
- The transaction-cost band exceeded the median basis by 12.7 times at its tightest.
- The two dollar legs opened the observed dislocations and the cross leg closed them.
- The basis is close to unforecastable from its own components.
- 31 July had a closure half-life of 1,003 seconds.

Not supported: a permanent regime change, a uniformly slower stress regime, a threefold widening of the intraday basis, or any executable dislocation.

Appendix: where each claim is written down

File	Claim it supports
00_daily_bias.csv	Seven days breach the bias gate; 1 and 2 August by fifteen-fold
00_reconciliation.csv	Tail agreement of each leg with its implied value
01_summary_stats.csv	Moments of the basis; excess kurtosis 277
01_closures.csv	Eight closures detected from the data
02_model_evidence.csv	Posterior over the four segmentations
02_null_calibration.csv	Behaviour of the estimator when no change exists
02_sensitivity.csv	Twenty specifications, all selecting two changepoints
03_var_fit.csv	Residual scales, correlations, sd of the daily mean basis
03_error_correction.csv	α , λ , HAC intervals, both half-lives
03_closure_speed.csv	Daily λ and half-life, 53 sessions
03_sensitivity.csv	Closure ratio across conventions, 1.5 to 24.8
04_attribution.csv	Opening and closing shares by identity
04_weak_exogeneity.csv	Sign restriction on α , all three regimes
04_basis_causality.csv	Effect sizes against placebo thresholds
05_execution.csv	Band width, dispersion, band-to-gap ratio
05_haircut.csv	Share clearing as a function of the spread scale
05_latency.csv, 05_edge_velocity.csv	Diagnostics on the simultaneity limit
06_normalisation.csv	Noise and persistence split of the level change
06_closure_reconciliation.csv	Five estimands of closure speed
06_boundary_audit.csv	Censoring at the rollover boundary
07_boundary_named.csv	The decomposition under each summarisation of the onset
07_boundary_sweep.csv	The decomposition at all 128 candidate boundaries
02_changepoint_posterior.png	Excursion, bimodal onset, fitted segment levels
03_closure_speed.png	Pooled stress mean set by one day
05_bands.png	Band against basis on a log axis
07_boundary_sweep.png	The decomposition against the regime boundary